

Distributional Reinforcement Learning via Moment Matching

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Abstract

We consider the problem of learning a set of probability distributions from the empirical Bellman dynamics in distributional reinforcement learning (RL), a class of state-of-the-art methods that estimate the distribution, as opposed to only the expectation, of the total return. We formulate a method that learns a finite set of statistics from each return distribution via neural networks, as in (Bellemare, Dabney, and Munos 2017; Dabney et al. 2018b). Existing distributional RL methods however constrain the learned statistics to *predefined* functional forms of the return distribution which is both restrictive in representation and difficult in maintaining the predefined statistics. Instead, we learn *unrestricted* statistics, i.e., deterministic (pseudo-)samples, of the return distribution by leveraging a technique from hypothesis testing known as maximum mean discrepancy (MMD), which leads to a simpler objective amenable to backpropagation. Our method can be interpreted as implicitly matching all orders of moments between a return distribution and its Bellman target. We establish sufficient conditions for the contraction of the distributional Bellman operator and provide finite-sample analysis for the deterministic samples in distribution approximation. Experiments on the suite of Atari games show that our method outperforms the distributional RL baselines and sets a new record in the Atari games for non-distributed agents.

Introduction

A fundamental aspect in reinforcement learning (RL) is the value of an action in a state which is formulated as the expected value of the *return*, i.e., the expected value of the discounted sum of rewards when the agent follows a policy starting in that state and executes that action (Sutton, Barto et al. 1998). Learning this expected action-value via Bellman’s equation (Bellman 1957) is central to value-based RL such as temporal-difference (TD) learning (Sutton 1988), SARSA (Rummery and Niranjan 1994), and Q-learning (Watkins and Dayan 1992). Recently, however, approaches known as *distributional* RL that aim at learning the distribution of the return have shown to be highly effective in practice (Morimura et al. 2010b,a; Bellemare, Dabney, and Munos 2017; Dabney et al. 2018b,a; Yang et al. 2019).

Despite many algorithmic variants with impressive practical performance (Bellemare, Dabney, and Munos 2017;

Dabney et al. 2018b,a; Yang et al. 2019), they all share the same characteristic that they explicitly learn a set of statistics of *predefined* functional forms to approximate a return distribution. Using predefined statistics can limit the learning due to the statistic constraints it imposes and the difficulty to maintain such predefined statistics. In this paper, we propose to address these limitations by instead learning a set of *unrestricted* statistics, i.e., deterministic (pseudo-)samples, of a return distribution that can be evolved into any functional form. We observe that the deterministic samples can be deterministically learned to simulate a return distribution by utilizing an idea from statistical hypothesis testing known as maximum mean discrepancy (MMD). This novel perspective requires a careful design of algorithm and a further understanding of distributional RL associated with MMD.

Leveraging this perspective, we are able to provide a novel algorithm to eschew the predefined statistic limitations in distributional RL and give theoretical understanding of distributional RL within this perspective. Our approach is also conceptually amenable for natural extension along the lines of recent modelling improvements to distributional RL brought by Implicit Quantile Networks (IQN) (Dabney et al. 2018a), and Fully parameterized Quantile Function (FQF) (Yang et al. 2019). Specifically, our key contributions are

1. We provide a novel approach to distributional RL using deterministic samples via MMD that addresses the limitations in the existing distributional RL that use predefined statistics;
2. We provide theoretical understanding of distributional RL within our framework, specifically the contraction property of the distributional Bellman operator and the non-asymptotic convergence of the approximate distribution from deterministic samples;
3. We demonstrate the practical effectiveness of our framework in both tabular RL and large-scale experiments where our method outperforms the standard distributional RL methods and even establishes a new record in the Atari games for non-distributed agents.

Outline. After carefully reviewing relevant background and related works, and discussing their limitations, we present our novel algorithmic approach to address these issues followed by theoretical analysis. We then present the experi-

ments to confirm the effectiveness of our approach empirically and conclude our work.

Background and Related Works

Expected RL

In a standard RL setting, an agent interacts with an environment via a Markov Decision Process $(\mathcal{S}, \mathcal{A}, R, P, \gamma)$ (Puterman 2014) where $\mathcal{S}$ and $\mathcal{A}$ denote state and action spaces, resp., R the reward measure, $P(\cdot|s, a)$ the transition kernel measure, and $\gamma \in [0, 1)$ a discount factor. A policy $\pi(\cdot|s)$ maps a state to a distribution over the action space.

Given a policy π , the discounted sum of future rewards following policy π is the random variable

$$Z^\pi(s, a) = \sum_{t=0}^{\infty} \gamma^t R(s_t, a_t), \quad (1)$$

where $s_0 = s, a_0 = a, s_t \sim P(\cdot|s_{t-1}, a_{t-1}), a_t \sim \pi(\cdot|s_t)$, and $R(s_t, a_t) \sim \mathcal{R}(\cdot|s_t, a_t)$. The goal in expected RL is to find an optimal policy π^* that maximizes the action-value function $Q^\pi(s, a) := \mathbb{E}[Z^\pi(s, a)]$. A common approach is to find the unique fixed point $Q^* = Q^{\pi^*}$ of the Bellman optimality operator (Bellman 1957) $T : \mathbb{R}^{\mathcal{S} \times \mathcal{A}} \rightarrow \mathbb{R}^{\mathcal{S} \times \mathcal{A}}$ defined by

$$TQ(s, a) := \mathbb{E}[R(s, a)] + \gamma \mathbb{E}_P[\max_{a'} Q(s', a')], \forall (s, a).$$

A standard approach to this end is Q-learning (Watkins and Dayan 1992) which maintains an estimate Q_θ of the optimal action-value function Q^* and iteratively improves the estimation via the Bellman backup

$$Q_\theta(s, a) \leftarrow \mathbb{E}[R(s, a)] + \gamma \mathbb{E}_P[\max_{a'} Q_\theta(s', a')].$$

Deep Q-Network (DQN) (Mnih et al. 2015) achieves human-level performance on the Atari benchmark by leveraging a convolutional neural network to represent Q_θ while using a replay buffer and a target network to update Q_θ .

Additional Notations

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be an open set. Let $\mathcal{P}(\mathcal{X})$ be the set of Borel probability measures on $\mathcal{X}$. Let $\mathcal{P}(\mathcal{X})^{\mathcal{S} \times \mathcal{A}}$ be the Cartesian product of $\mathcal{P}(\mathcal{X})$ indexed by $\mathcal{S} \times \mathcal{A}$. For any $\alpha \geq 0$, let $\mathcal{P}_\alpha(\mathcal{X}) := \{p \in \mathcal{P}(\mathcal{X}) : \int_{\mathcal{X}} \|x\|^\alpha p(dx) < \infty\}$. When $d = 1$, let $m_n(p) := \int_{\mathcal{X}} x^n p(dx)$ be the n -th order moment of a distribution $p \in \mathcal{P}(\mathcal{X})$, and let

$$\mathcal{P}_*(\mathcal{X}) = \left\{ p \in \mathcal{P}(\mathcal{X}) : \limsup_{n \rightarrow \infty} \frac{|m_n(p)|^{1/n}}{n} = 0 \right\}.$$

Note that if $\mathcal{X}$ is a bounded domain in $\mathbb{R}$, then $\mathcal{P}_*(\mathcal{X}) = \mathcal{P}(\mathcal{X})$. Denote by δ_z the Dirac measure, i.e., the point mass, at z . Denote by Δ_n the n -dimensional simplex.

Distributional RL

Instead of estimating only the expectation Q^π of Z^π , distributional RL methods (Bellemare, Dabney, and Munos 2017; Dabney et al. 2018b,a; Rowland et al. 2018; Yang et al. 2019) explicitly estimate the return distribution $\mu^\pi =$

$\text{law}(Z^\pi)$ as an auxiliary task. Empirically, this auxiliary task has been shown to significantly improve the performance in the Atari benchmark. Theoretically, in the policy evaluation setting, the distributional version of the Bellman operator is a contraction in the p -Wasserstein metric (Bellemare, Dabney, and Munos 2017) and Cr  mer distance (Rowland et al. 2018) (but not in total variation distance (Chung and Sobel 1987), Kullback-Leibler divergence and Komogorov-Smirnov distance (Bellemare, Dabney, and Munos 2017)). The contraction implies the uniqueness of the fixed point of the distributional Bellman operator. In control settings with tabular function approximations, distributional RL has a well-behaved asymptotic convergence in Cr  mer distance when the return distributions are parameterized by categorical distributions (Rowland et al. 2018). Bellemare et al. (2019) establish the asymptotic convergence of distributional RL in policy evaluation in linear function approximations. Lyle, Bellemare, and Castro (2019) examine behavioural differences between distributional RL and expected RL, aligning the success of the former with non-linear function approximations.

Categorical distributional RL (CDRL) CDRL (Bellemare, Dabney, and Munos 2017) approximates a distribution η by a categorical distribution $\hat{\eta} = \sum_{i=1}^N \theta_i \delta_{z_i}$ where $z_1 \leq z_2 \leq \dots \leq z_N$ is a set of fixed supports and $\{\theta_i\}_{i=1}^N$ are learnable probabilities. The learnable probabilities $\{\theta_i\}_{i=1}^N$ are found in such way that $\hat{\eta}$ is a projection of η onto $\{\sum_{i=1}^N p_i \delta_{z_i} : \{p_i\}_{i=1}^N \in \Delta_N\}$ w.r.t. the Cr  mer distance (Rowland et al. 2018). In practice, C51 (Bellemare, Dabney, and Munos 2017), an instance of CDRL with $N = 51$, has shown to perform favorably in Atari games.

Quantile regression distributional RL (QRDRL) QRDRL (Dabney et al. 2018b) approximates a distribution η by a mixture of Diracs $\hat{\eta} = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i}$ where $\{\theta_i\}_{i=1}^N$ are learnable in such a way that $\hat{\eta}$ is a projection of η on $\{\frac{1}{N} \sum_{i=1}^N \delta_{z_i} : \{z_i\}_{i=1}^N \in \mathbb{R}^N\}$ w.r.t. to the 1-Wasserstein distance. Consequently, $\theta_i = F_\eta^{-1}(\frac{2i-1}{2N})$ where F_η^{-1} is the inverse cumulative distribution function of η . Since the quantile values $\{F_\eta^{-1}(\frac{2i-1}{2N})\}$ at the fixed quantiles $\{\frac{2i-1}{2N}\}$ is a minimizer of an asymmetric quantile loss from quantile regression literature (thus the name QRDRL) and the quantile loss is compatible with stochastic gradient descent (SGD), the quantile loss is used for QRDRL in practice. QRDQN-1 (Dabney et al. 2018b), an instance of QRDRL with Huber loss, performs favorably empirically in Atari games.

Other distributional methods Some recent distributional RL methods have made modelling improvements to QRDRL. Two typical improvements are from Implicit Quantile Networks (IQN) (Dabney et al. 2018a), and Fully parameterized Quantile Function (FQF) (Yang et al. 2019). IQN uses implicit models to represent the quantile values $\{\theta_i\}$ in QRDRL, i.e., instead of being represented by fixed network outputs, $\{\theta_i\}$ are the outputs of a differentiable function (e.g., neural networks) on the samples from a base sampling distribution (e.g., uniform). FQF further improves IQN by optimizing the locations of the base samples for IQN, instead

of using random base samples as in IQN, i.e., both quantiles and quantile values are learnable in FQF.

Predefined statistic principle

Formally, a statistic is any functional $\zeta : \mathcal{P}(\mathcal{X}) \rightarrow \mathbb{R}$ that maps a distribution $p \in \mathcal{P}(\mathcal{X})$ to a scalar $\zeta(p)$, e.g., the expectation $\zeta(p) = \int_{\mathcal{X}} xp(dx)$ is a common statistic in RL. Here, we formally refer to a *predefined statistic* as the one whose functional form is specified before the statistic is learned. In contrast, an *unrestricted statistic* does not subscribe to any specific functional form (e.g., the median of a distribution η is a predefined statistic as its functional form is predefined via $F_{\eta}^{-1}(\frac{1}{2})$ while any empirical sample $z \sim \eta$ can be considered an unrestricted statistic of η).

Though CDRL and QRDL are two different variants of distributional RL methodology, they share a unifying characteristic that they both explicitly learn a finite set of predefined statistics, i.e., statistics of *predefined* functional forms (Rowland et al. 2019). We refer to this as predefined statistic principle. This is clear for QRDL as the statistics to be learned about a distribution η are $\{\zeta_1, \dots, \zeta_N\}$ where

$$\zeta_i(\eta) := F_{\eta}^{-1}\left(\frac{2i-1}{N}\right), \forall i \in \{1, \dots, N\}.$$

It is a bit more subtle for CDRL. It can be shown in (Rowland et al. 2019) that CDRL is equivalent to learning the statistics $\{\zeta_1, \dots, \zeta_{N-1}\}$ where

$$\zeta_i(\eta) := \mathbb{E}_{Z \sim \eta} \left[1_{\{Z < z_i\}} + 1_{\{z_i \leq Z < z_{i+1}\}} \frac{z_{i+1} - Z}{z_{i+1} - z_i} \right], \forall i.$$

Learning predefined statistics as in CDRL and QRDL however can suffer two limitations in (i) statistic representation and (ii) difficulty in maintaining the predefined statistics. Regarding (i), given the same fixed budget of N statistics to approximate a return distribution η , CDRL restrictively associates the statistic budget to N fixed supports $\{z_i\}_{i=1}^N$ while QRDL constrains the budget to N quantile values at specific quantiles. Instead, the statistic budget should be freely learned into any form as long as it could simulate the target distribution η sensibly. Regarding (ii), the fixed supports in CDRL require a highly involved projection step to be able to use KL divergence as the Bellman backup changes the distribution supports; QRDL requires that the statistics must satisfy the constraints for valid quantile values at specific quantiles, e.g., the statistics to be learned are order statistics. In fact, QR-DQN (Dabney et al. 2018b), a typical instance of QRDL, implicitly maintains the order statistics via an asymmetric quantile loss but still does not guarantee the monotonicity of the obtained quantile estimates. A further notice regarding (ii) recognized in (Rowland et al. 2019) is that since in practice we do not observe the environment dynamic but only samples of it, a naive update to learn the predefined statistics using such samples can collapse the approximate distribution due to the different natures of samples and statistics (in fact, (Rowland et al. 2019) proposes imputation strategies to overcome this problem). Instead, the statistics to be learned should be free of all such difficulties to reduce the learning burden.

One might say that IQN/FQF (discussed in the previous subsection) can help QRDL overcome these limitations. While the modeling improvements in IQN/FQF are practically effective, IQN/FQF however still embrace the predefined statistic principle above as they built upon QRDL with an improved modelling capacity. In this work we propose an alternative approach to distributional RL that directly eschews the predefined statistic principle used in the prior distributional RL methods, i.e., the finite set of statistics in our approach can be evolved into any functional form and thus also reduces the need to maintain any statistic constraints. If we informally view improvements to CDRL/QRDL into two dimensions: either modelling dimension or statistic dimension, IQN/FQF lie in the modelling dimension while our work belongs to the statistic dimension. We notice that this does not necessarily mean one approach is better than the other, but rather two orthogonal approaches where the modelling improvements in IQN/FQF can naturally apply to our work to further improve the modeling capacity. We leave these modeling extensions to the future work and focus the present work only on unrestricted statistics with the simplest modelling choice as possible.

Distributional RL via Moment Matching

Maximum mean discrepancy

Let $\mathcal{F}$ be a reproducing kernel Hilbert space (RKHS) associated with a continuous kernel $k(\cdot, \cdot)$ on $\mathcal{X}$. Consider $p, q \in \mathcal{P}(\mathcal{X})$, and let Z and W be two random variables with distributions p and q , respectively. The maximum mean discrepancy (MMD) (Gretton et al. 2012) between p and q is defined as

$$\begin{aligned} \text{MMD}(p, q; \mathcal{F}) &:= \sup_{f \in \mathcal{F}: \|f\|_{\mathcal{F}} \leq 1} (\mathbb{E}[f(Z)] - \mathbb{E}[f(W)]) \\ &= \|\psi_p - \psi_q\|_{\mathcal{F}} \\ &= \left(\mathbb{E}[k(Z, Z')] + \mathbb{E}[k(W, W')] - 2\mathbb{E}[k(Z, W)] \right)^{1/2} \end{aligned}$$

where $\psi_p := \int_{\mathcal{X}} k(x, \cdot) p(dx)$ is the Bochner integral, i.e., the mean embedding of p into $\mathcal{F}$ (Smola et al. 2007), and Z' (resp. W') is a random variable with distribution p (resp. q) and is independent of Z (resp. W). In sequel, we interchangeably refer to MMD by $\text{MMD}(p, q; \mathcal{F})$, $\text{MMD}(p, q; k)$, or $\text{MMD}(p, q)$ if the context is clear.

Empirical approximation. Given empirical samples $\{z_i\}_{i=1}^N \sim p$ and $\{w_i\}_{i=1}^M \sim q$, MMD admits a simple empirical estimate as

$$\begin{aligned} \text{MMD}_b^2(\{z_i\}, \{w_i\}; k) &= \\ \frac{1}{N^2} \sum_{i,j} k(z_i, z_j) &+ \frac{1}{M^2} \sum_{i,j} k(w_i, w_j) - \frac{2}{NM} \sum_{i,j} k(z_i, w_j). \end{aligned}$$

Though there is also a simple unbiased estimate of MMD, the *biased estimate* MMD_b has smaller variance in practice and thus is adopted in our work.